

Module - 2

Materials for Electronic Applications

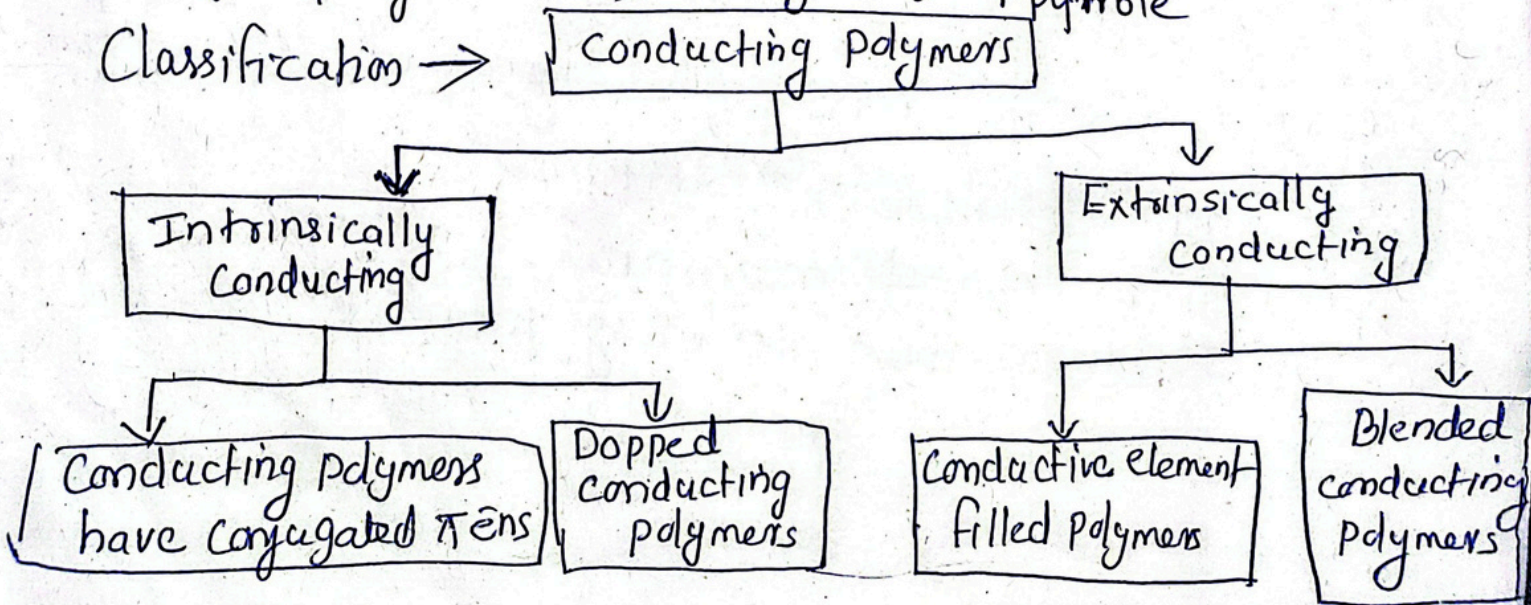
Polymers

Conducting polymers

A polymer which can conduct electricity is termed as ~~conducting~~ polymers.

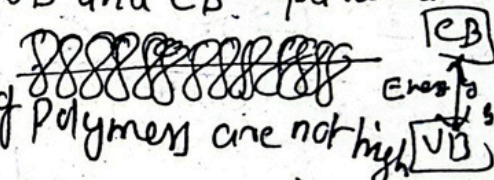
eg: poly aniline, poly ~~pyrrole~~ etc.
 $\uparrow$ pyrrole

Classification $\rightarrow$



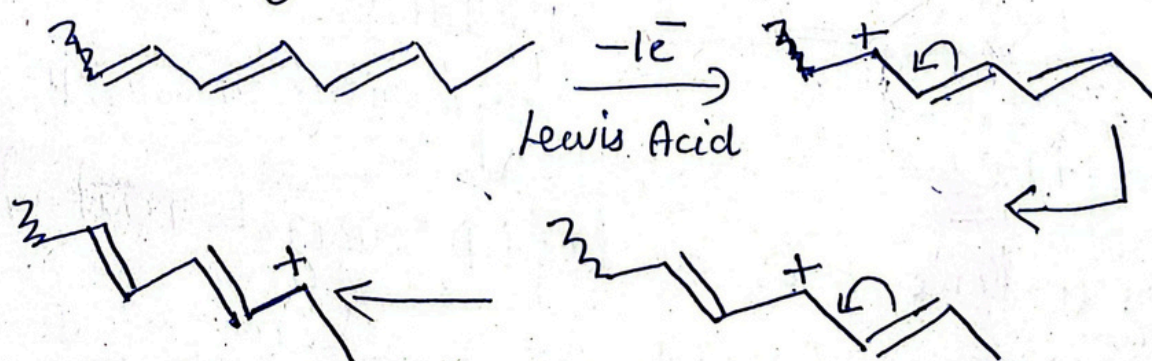
Conducting polymers have conjugated π electrons

Here the polymers have conjugated double bonds along the backbone of the polymer. The orbitals of conjugated π electrons overlap over the entire backbone of the polymer, resulting in the formation of valence band and conduction band. Electrical conduction would occur when electron from valence band are excited to conduction band either thermally ~~thera~~ or photolytically. Since VB and CB separated by a significant band gap, conductivity of ~~polymers~~ Doped conducting polymers are not high.



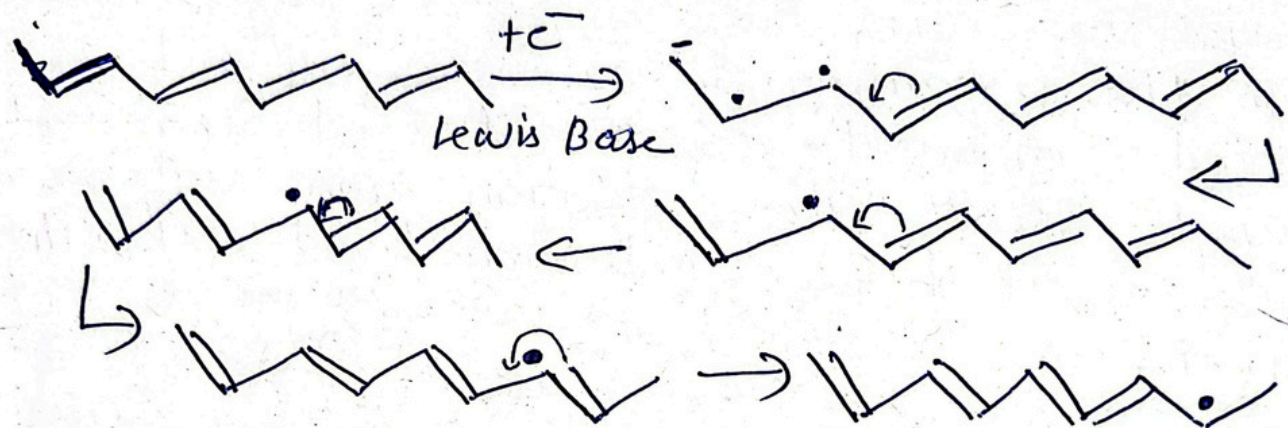
Conductivity of a polymer can be increased by creating positive or negative charge on polymer backbone by oxidation or reduction. This process is termed as doping.

P-doping: It is done by oxidation process. The polymer is treated with Lewis acid. Then holes ($\oplus$ charge) are created and can move along the polymer chain.



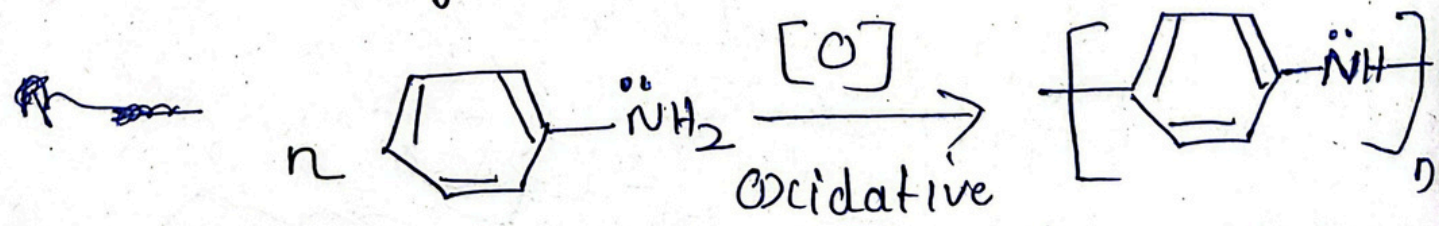
Propagation of polaron through a conjugated polymer chain by shifting of double bonds.

n-doping - is done by reduction process.
 Here electrons are introduced into the polymer chain having conjugated double bonds. The reduction of polyacetylene by a Lewis base leads to the formation of Polaron. $\longleftrightarrow$



Poly aniline

Prepared by Electrochemical method.
 It is done by dipping Platinum electrodes in aqueous solution of aniline and a potential of 0.7V is applied, polyaniline is formed on the surface of positive electrode



Properties

polymerisation + n H₂O

- Polyaniline is a light weight material
- It possess very high chemical resistance.
- Depend polyaniline is a good conductor comparable to copper

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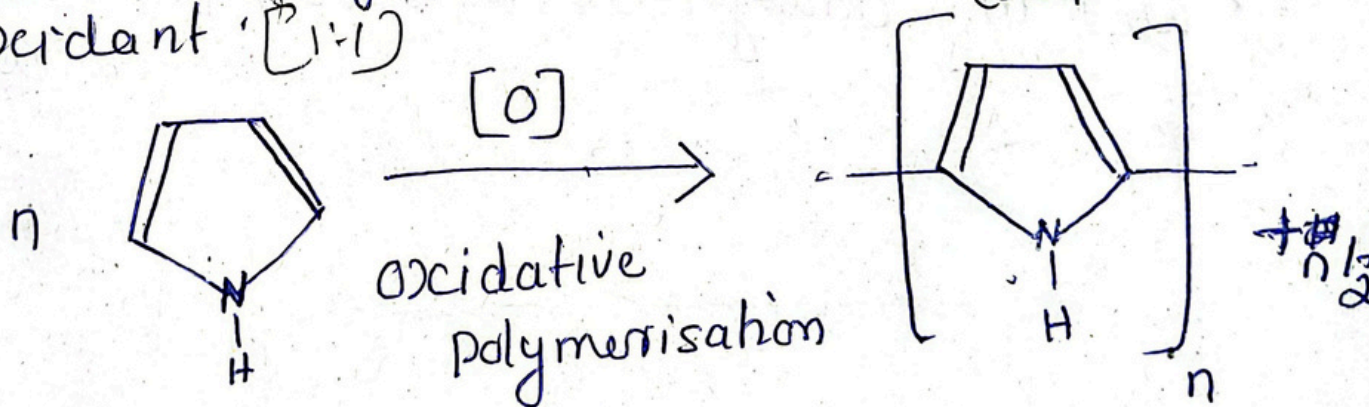
Polyaniline shows electrochromism

Applications

- Use as Electrode material for rechargeable batteries
- used in biosensors and solar energy absorption
- used in printer circuit board
- used to make light weight rechargeable batteries

Poly Pyrrole

Poly Pyrrole can be prepared by chemical oxidative polymerisation of pyrrole using ammonium persulphate as oxidant (1:1) (sulphate)



[Chemical oxidative polymerisation can be carried out in a beaker by mixing 0.1 mg solution of pyrrole and 0.1 M ammonium persulphate)

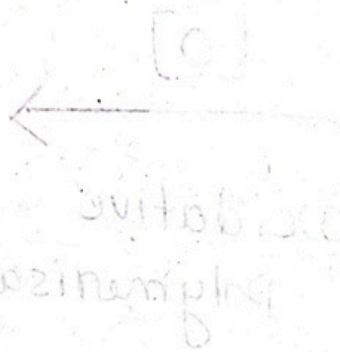
Properties

9 5

- Poly Pyrrole has excellent electrical, thermal and mechanical properties.
- It has good environmental stability and higher conductivity.
- Poly pyrrole films are yellow but darken in air due to oxidation.
- It has very high chemical resistance.
- They are amorphous and give weak diffraction.

Applications

- Widely used in biosensors and immunosensors.
- used in smart windows and displays.
- used in light weight rechargeable batteries and electronic devices.
- It have applications in tissue engineering.
- used in analytical sensors, photovoltaic devices.

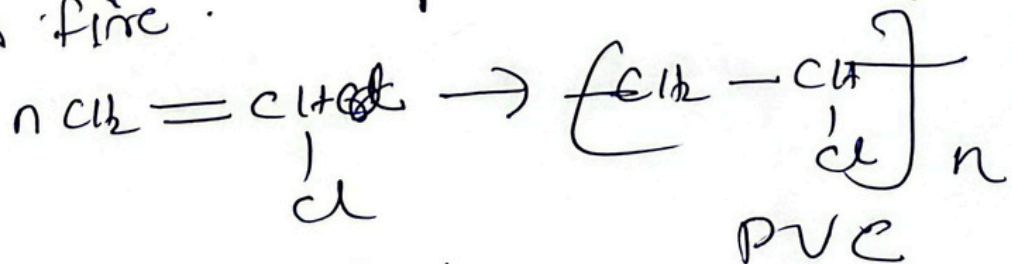


Fire-retardant polymers

Fire retardant polymers resist the ignition and slow the spread of flames. They have the ability to slow the spread of fire. eg: Novolac, ~~Bakelite~~ Bakelite, melamine formaldehyde; ~~silicon~~ polymers.

Halogenated fire retardant polymers: This class of polymers includes ~~halogenated~~ brominated compounds or

Halogenated fire retardant polymers include brominated compounds or chlorinated compounds. In ~~the case of~~ Halogenated polymers introduced halogen radical into the fire, which combines with $H^\bullet$ radical and $OH^\bullet$ ^{in fire} to form stable molecules like HX or HOX resulting in quenching of ~~characteristic~~ fire.

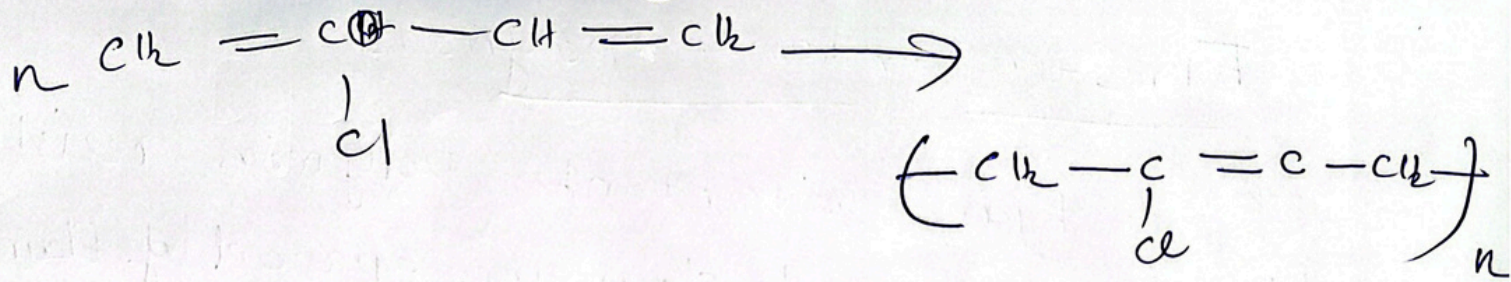


eg: PVC

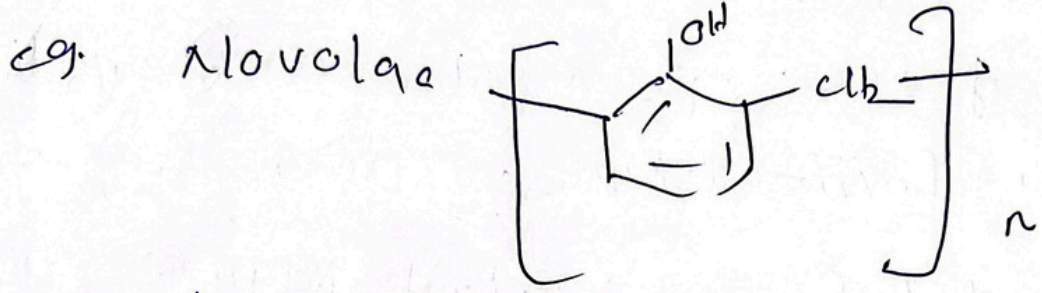


brominated
butyl rubber.

⑧



Non Halogenated Fire-retardant ^{neoprene.} polymers



c: Bakelite.



Module - 2
Materials for Electronic Applications
Nanomaterials

Nanomaterials

Sig. R

Nanomaterials are materials whose characteristic length scale lies within the nanometric range ($1-100\text{ nm}$) at least in one dimension

Classification of Nanomaterials Based on dimension

1. Zero-dimension (0-D) :- Nanomaterials in which all dimensions are measured within the nanoscale. eg: Electron is confined in a 3-D space. **Nano particles.**
2. One-dimension - Nanomaterials :- are materials in which two dimensions are in nanoscale. eg: Nano wires, Nanofibers, Nanotubes.
3. Two-dimension Nano materials :- are materials in which only one dimension is in nanometric range. eg: Nano films, Nano layers, and Nano coatings.
4. Three-dimension Nanomaterials :- Here all the dimensions are outside the nanometric range. Bundles of nanowires and tubes.

Classification based on Materials

(1) Carbon-based nanomaterials:- These are defined as materials in which the nanocomponent is pure carbon. OR. nano materials of which main component is pure carbon
eg: Carbon nano tubes, Fullerenes

(2) Metal based Nanomaterials:- Metal based nanomaterials are materials made of metallic nanoparticles like Silver, gold and metal oxides

(3) Nano Composites:- (Composite nanomaterials contain a mixture of simple nanoparticles within a bulk material.) Nano Particles provide better Physical, mechanical and Chemical Properties to the initial bulk material.

(4) Nano-Polymers or Dendrimers:- Dendrimers are nano sized polymers built from branched units. These are tree like molecules with defined cavities. They can be functionalised at the surface and can hide molecules in their cavities. They are widely used in drug delivery.

Biological Nanomaterials:- These

nanomaterials are biological origin. The important features of these particles are

- (1) Self assembly properties
 - (2) Specific molecular recognition
- eg: DNA - nanoparticles

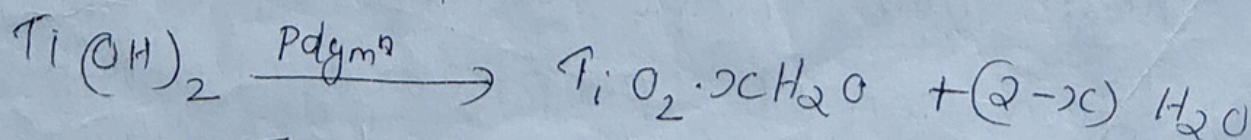
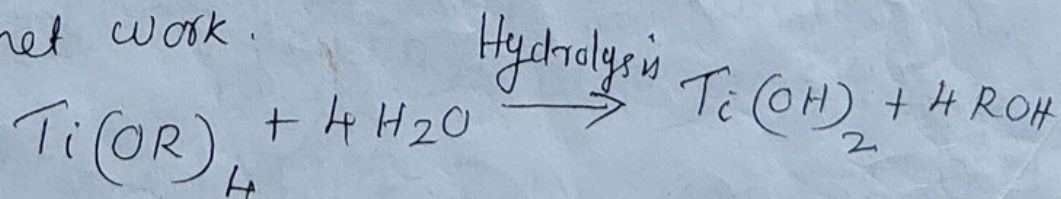
Synthesis of Nanomaterials

(preparation) [chemical methods of preparation]

(1) Hydrolysis Method

Nanoparticles of metal oxides can be prepared by the hydrolysis of their alkoxide solutions under controlled conditions.

eg: Titanium alkoxide get hydrolysed and polymerised to form a three dimensional oxide net work.



The precipitate so obtained was

washed with ethanol and dried at 100°C in a vacuum system for 3 hrs. It is then subjected to annealing process get the nanoparticles.

(2) Sol - Gel Method: is based on the

Phase transformation of a sol into a gel.

A sol is colloidal system of nano-solid particles dispersed in a liquid. A gel is colloidal system in which liquid droplets are dispersed in a network of solid nanoparticles.

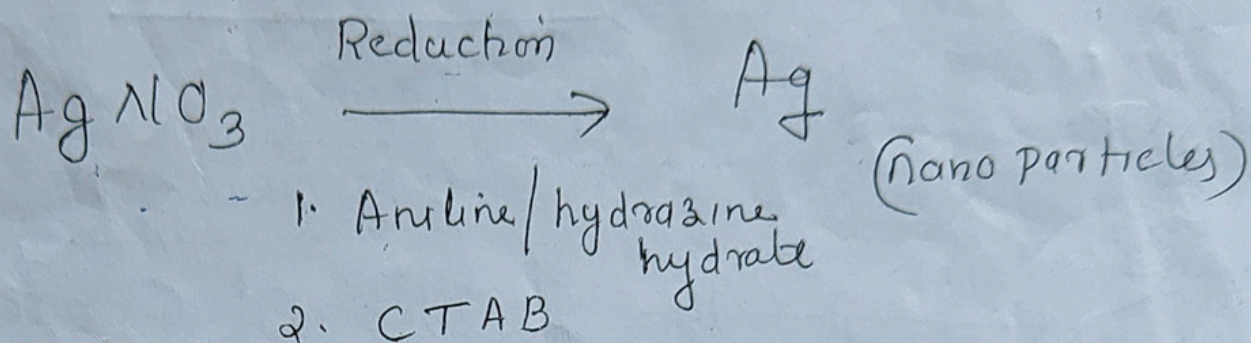
Hydrolysis of metal alkoxides can give a sol at a suitable temperature and pH:

[The sol contains impurities. In order to remove impurities, ~~the sol is the sol~~

the sol is transformed into a gel by changing the pH or other factors. The gel can be purified by filtration and washing with suitable solvents. The purified gel on drying give solid nanoparticles.]

Reduction Nano particles of gold and silver can be prepared by the reduction of their respective solutions using

Reducing agents such as sodium borohydride, hydrazine hydrate, Aniline solution^{etc} along with a protective agent.



Electro reduction

Copper nanoparticles have been prepared by electroreduction process using copper plating bath containing acidified copper sulphate solution. The spongy nanoparticles are formed at the cathode. The spongy layers of copper can be separated to give fine particles.

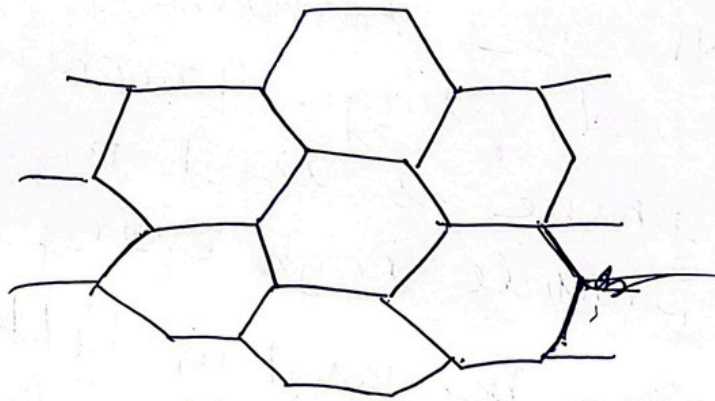
Applications

- (1) Carbon nanotubes used in transistors.
- (2) Graphene sheets are used as electrodes in ultra capacitors
- (3) carbon nanotube are used for making paper batteries
- (4) Nano particles are used for the treatment of tumors.

Graphene

(1)

Graphene is an allotrope of carbon consisting of monolayer of carbon atoms arranged in a two dimensional honey comb lattice. Graphene is a two dimensional monolayer of ^{carbon} atoms. ~~thickness of 20~~
Structure



Honey Comb lattice of graphene

Properties

- It is considered to be the lightest, thinnest, strongest material that conducts heat and electricity.
- Stronger than diamond and ten times more ~~the~~ conducting than copper.
- Graphene has high electrical conductivity.
- They possess high elastic modulus and strength. It is 200 times stronger than steel.

- It is chemically ^② inert.
- It has high surface area due to its single atomic thickness and hence useful for applications in energy storage devices.

Applications

- It is used in touch screens.
- Small size transistors can be developed using graphene.
- Graphene is used in supercapacitors to provide high energy density.
- It can be used in filters to remove pollutants and toxins from the air.

Carbon nano tubes

Graphene sheet is rolled up into a cylindrical shape is called carbon nanotube.

Types of Carbon nanotubes, (based on no. of layers)

Single walled Carbon nanotubes.

In single, ^{walled} carbon nanotubes, only one graphene sheet is used to form the cylindrical tubes.

(3)

Graph multilayered Carbon is a stack of graphene sheets rolled up into concentric cylinders.

Properties

- (1) They possess remarkable tensile strength.
- (2) Thermally stable upto 2800°C in vacuum and about 750°C in air.
- (3) Electrical conductivity is better than metals.
- (4) They are strongest and stiffest materials.

Applications

- (1) used in electrical circuits
- (2) used for the manufacture of electronic devices like Paper Batteries.
- (3) used in solar cells and supercapacitors.
- (4) used in aerospace and automotive industry due to its high strength, stiffness and reduced weight.

(4)

Carbon Quantum dots ~~QDs~~ (CQDs)

Carbon Quantum dots are zero dimensional carbon nanoparticles. (It consists of ultrafine - quasi-spherical Carbon nanoparticles with size less than 10 nm.)

Properties They have high biocompatibility and good dispersibility. They exhibit strong fluorescence and photoluminescence and used in imaging.

2) They show chemiluminescence and used in sensing applications.

4) CQDs show excellent photo stability.

Applications

① Used in fluorescent bio-imaging

② They used in drug delivery system.

3) Lighting - used in LED materials.

Fullerenes: are class of cage like carbon compounds composed of fused pentagonal or hexagonal sp^2 carbon rings, eg. C_{60} C_{70} .

C_{60} contains 60 carbon atoms composed of fused pentagonal and hexagonal carbon rings. It contains 12 five membered rings and 20 six membered rings. It is also known as soccer football, or bucky ball. In fullerene carbon atoms are sp^2 and properties Fullerenes are sp^3 hybridised.

- ① Extremely strong molecules, and able to resist great pressure. ~~It is~~ so fullerenes are become harder than steel and diamond.
- ② Fullerenes are soluble in solvents like ~~total~~ toluene and CS_2 .
- ③ Light absorption. Fullerenes absorb strongly in the UV and moderately in the visible regions of the spectrum.

Applications

(1) coatings and lubricant

Fullerenes are used for making high performance coatings that resist wear and corrosion.

2. C_{60} can be used as excellent microscopic ball bearings, lubricant and catalyst.
3. Fullerenes can be used ~~as~~ in imaging techniques such as MRI or ultrasound.
4. Fullerenes are used as energy storage devices due to their electrical conductivity.
5. They are used in water treatment for the removal of pollutants and contaminants.

Module - 2
Materials for Electronic Applications
Organic Electronic Materials and
Devices

Dye - Sensitized Solar cells [DSSCs]

A dye sensitized solar cell is a Photoelectrochemical cell, which converts solar light absorbed by the dye to electric energy. It is a thin film solar cell.

Construction of Dye sensitised solar cells

It consists of

- (1) Transparent anode TiO_2
- (2) Dye sensitised nano particles: The dye absorb sunlight and inject photoexcited electrons to conduction band.
- (3) Electrolyte - Redox couple of Iodide and Triiodide is used as electrolyte I^-/I_3^-
- (4) Counter electrode - Platinum electrode act as cathode.

Working

The sunlight passes through the transparent anode and electrons of the dye molecule then excited and are injected into the conduction band of TiO_2 semi conductor.

(7)

The e^- s travels from metal oxide to cathode (Pt) through external circuit and flow into the Iodide electrolyte and generate electricity.

The oxidised dye molecules

get reduced by gaining e^- s from the electrolyte, and come back to the original form.

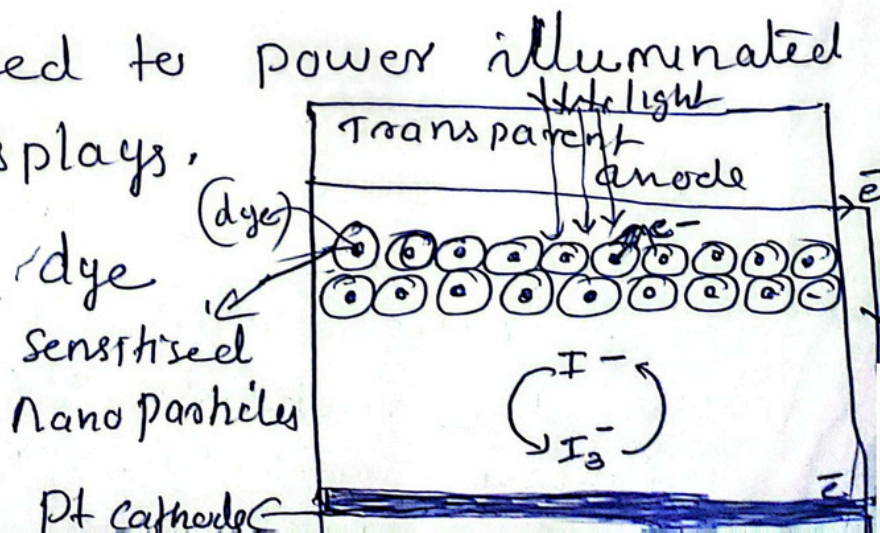
The oxidised electrolyte move towards the cathode and get reduced. This process is continues to produce current.

Advantages

- (1) They are light weight and find applications in vehicles.
- (2) Natural pigment and organic dyes are employed which are eco-friendly.

Applications

- used in electronic devices such as smart phones
- It can be used to power illuminated signs and displays.



(1)

Organic Light Emitting Diode

[OLED]

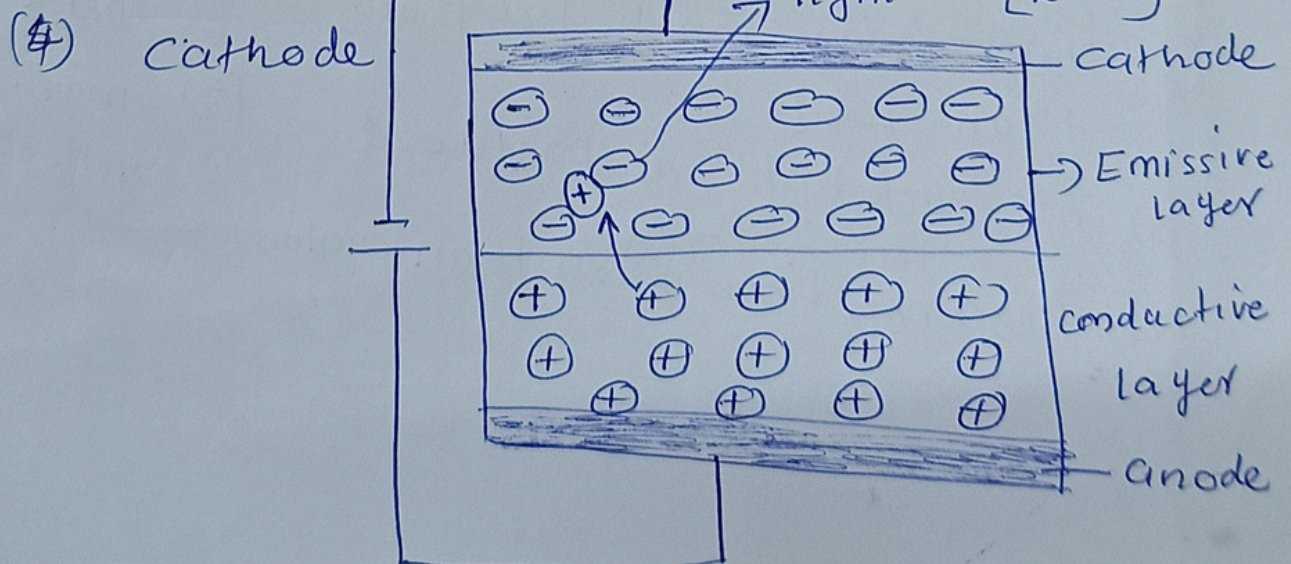
An organic light-emitting diode (OLED) is a light emitting diode in which the emissive organic layer emits light in response to an electric current.

An OLED structure consists of following parts.

(1) Anode - which adds holes when a current flows through the device

(2) Conductive layer: made of polyaniline and ~~known~~ known as hole transport layer [HTL]

(3) Emissive layer: Organic polymer known as an electron transport layer [ETL]



Fig

(2)

Working

To make an OLED light up, a Potential difference is applied across the anode and cathode. As the electricity starts to flow, the cathode receives electrons from the power source and the anode loses them. So cathode and emissive layer becomes negatively charged while anode and conductive layer becomes positively charged. Positive holes are much more mobile than negative electrons so they jump from conductive layer to emissive layer. Combination of a hole and electron take place and release energy in the form of light.

Advantages

- (1) OLEDs are brighter than LED
- (2) They consume much less power than LED
- (3) No backlight is produced
- (4) Response time is less than 0.01ms

Applications

- (1) OLED-TV
- (2) Mobile Phones with OLED screens
- (3) Smart watch with OLED screens

Materials⁽⁴⁾ in Quantum Computing Technology

(1) Spintronics or spin electronics

Spintronics is a field of electronics that utilizes the spin of electron, in addition to its charge, to process information in ~~new~~ generation transistors, lasers and integrated magnetic sensors. They act by storing information in electron spins, and transmitting this information via mobile electrons. Spin-orientation lasts longer than charge-based transformation, making them suitable for memory storage, magnetic sensors and Quantum computing.

Spintronics is basically connected to magnetism because the magnetic moment of electron is proportional to its spin. Spintronics make use of the magnetoresistive effects for sensing and storing information.

eg: Metal oxides such as MgO , Al_2O_3 , SrTiO_3 are the examples of metal based spintronics materials.

Applications

1. used in Magnetic Storage Devices
eg. $\left[\begin{array}{l} \text{Magnetic-random access} \\ \text{[MRAM] memory} \end{array} \right]$

2. Spintronic Transistors: — ~~These~~

3. Used in memory storage devices, magnetic sensors and Quantum Computing.

Super capacitors

Super capacitors or Ultra Capacitors are energy storage devices having high capacity with a capacitance value much higher than solid-state capacitors but with lower voltage limits. In super-capacitors energy is stored electrostatically and electrochemically.

A super capacitor consists of

- (1) electrodes $\Rightarrow$ Carbon nano tubes, graphene, CQDs are used as electrodes.
- (2) electrolyte $\Rightarrow$ tetraalkyl ammonium or Li-salt act as solid electrolyte.
- (3) separator $\Rightarrow$ Electrolyte membrane will act as separator. It prevents short-circuiting between electrodes.
- (4) collector $\Rightarrow$ Electrons are collected at the collector.
eg: metals like Al, Pt, Cu, Ni act as collectors.

Advantages of ⁽¹¹⁾ Supercapacitors

1. Longer service life compared to the battery
2. Low resistance which enables them to produce high load currents.
3. High efficiency, higher charging rates and high power density.
4. Environmental Friendly.

Applications of Supercapacitors.

- (1) used in hybrid buses in combination with the battery to increase battery life and decrease the size.
 - (2) Memory devices in laptops, smart phones etc.
 - (3) used in solar powered road and farming devices, street light, wireless security camera etc.
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